



DEALER BEST PRACTICE SERIES

In-Process Staging Strategy

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

Component Rebuild Center (CRC) parts and component staging is critical to several facets of CRC operations. Staging is non-value adding and is typically unplanned or not maintained. Rebuild product should be staged between processing steps in a manner that consumes minimal resources (space, labor, equipment). Staging begins with major component receipt and ends with major component release to the customer, parts department, etc. Effective part/component staging reduces staging space requirements and technician rebuild labor with simpler storage equipment and product accessibility.

An effective staging strategy enables consistency and minimizes “lost” in-process product. Last-minute customer changes are less costly to re-schedule due to greater process flexibility. Properly used, an enhanced staging strategy provides increased capacity, efficient parts flow, and better space utilization.

2.0 Best Practice Description

For the scope of this Best Practice, parts/components staging begins after the major component is processed in the general disassembly/cleaning area(s).

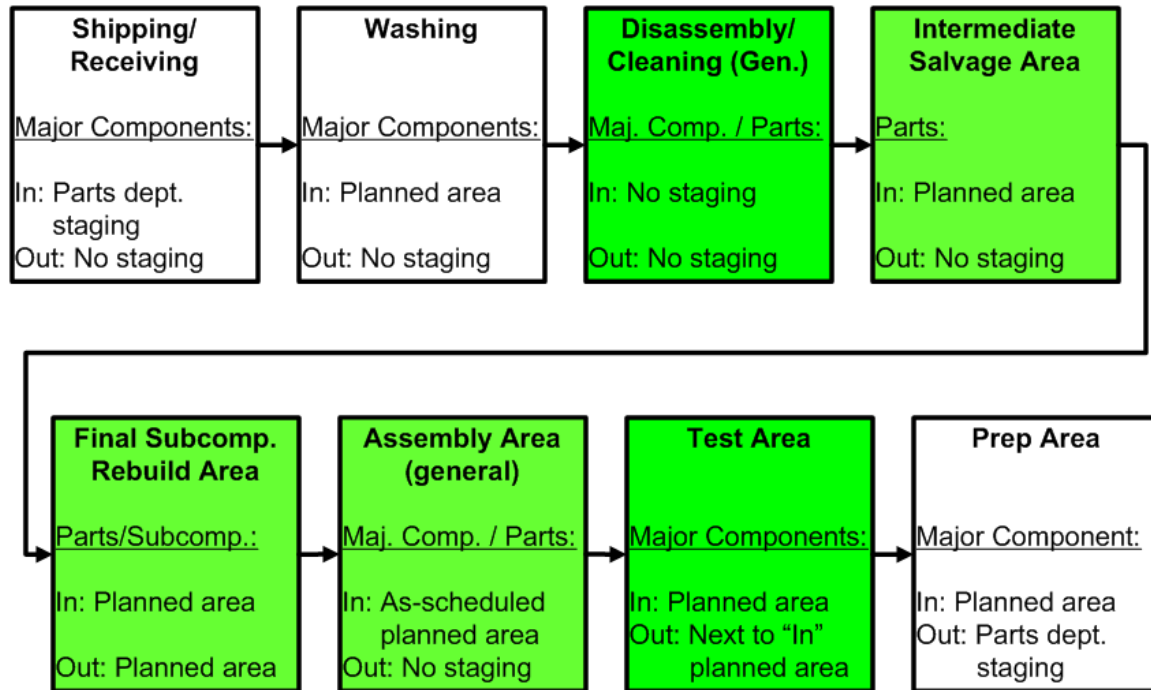
- Current practice is typically ineffective and inconsistent.
 - Once cleaned the product often stays in the cleaning area, (or wherever there is space) until it is sent to one of several component rebuild areas (CRA). The CRA may be specific to shafts, pumps & drives, front/rear covers, turbochargers, fuel systems, etc.
 - Once the CRA has completed processing the product it may be sent to wait in the general assembly bay until all of the parts/components are available for the assembly process.
 - Parts/components are often in one of several places, depending on who processed them and where the shop has space to park them. They may end up in an aisle traffic lane, close to a shop access door, or in any number of open places vulnerable to dirt, debris, water, and contact damage.
- An enhanced Staging Strategy is efficient and parts-friendly.
 - Once disassembled and cleaned, the parts/components must leave the cleaning area and move to the part/component rebuild area. The sooner the clean parts leave this area, the cleaner they stay.

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CRC Staging Strategy - CR Process Map (high-level)

Best Practice Scope included in green background



- Intermediate Salvage Area**

Includes machine shop, head disassembly/cleaning bay etc. Once processed, the parts are immediately sent to the incoming staging area of the next processing bay (such as water pump rebuild, head, or turbo rebuild bays). Do not stage completed work in an out-going staging area.

- Final Subcomponent Rebuild Bay**

Includes cooler assembly bay, head rebuild bay, etc. Once processed, the parts are staged in an outgoing staging area. This staging area is best located close to the incoming staging area, but it would be acceptable to locate it in a protected satellite area.

Eliminating the **General Cleaning Area** and intermediate salvage area's outgoing staging areas reduces parts contamination and parts wait time.

It is easier to protect the parts/components with specialized staging equipment.

Establishing the **Final Subcomponent Rebuild Bay** incoming and outgoing staging areas provides like-component staging/handling equipment and:

- Reduces staging space overall because, for example, many crankshafts can be stored efficiently with specialized equipment.
- Uses less costly space than is used in the major component assembly areas (smaller cranes, lower ceilings, etc.)

Delaying the **Major Component Assembly Bay** incoming staging until assembly occurs:

- Maintains assembly job flexibility. The bays are available for any job until one actually begins (since no parts are delivered until needed).
- Allows a minimally sized assembly bay to be more productive if the parts and subassemblies can be delivered as needed instead of all-at-once (congesting the bay).

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- **Major Component (General) Assembly Bays**
 - Do not stage parts/components in the bay until the major component is being assembled.
 - Do not stage assembled major components in an assembly bay out-going staging area. Eliminating the general assembly area's outgoing staging areas reduces transfer wait time for the test area and shop component storage area.
- **Test Cells**
Includes dynamometer or hydraulic test bench cell areas. Once processed, the parts are staged in an out-going staging area. This staging area is best located close to the incoming staging area, but it would be acceptable to locate it in a protected satellite area.

3.0 Implementation Steps

1. Document as-is staging methods and performance.
2. Define future staging strategy based on expected product type, volume, size, etc.
 - a. Plan for typical rebuild volumes as well as expected peak production business cycles.
 - b. Investigate specialized product storage (shelves, carts, racks, etc.) and handling (cranes, carts, lift trucks, tugs, etc) equipment.
 - c. Use Cat Facility Development and shop equipment supplier expertise as needed.
3. Apply future strategy to shop layout plans.
 - a. Consider staging location's effects on product protection and flow paths.
 - b. Add changes to shop layout drawing.
 - c. Include shop technician teams in the review/application stages.
 - d. Use Cat Facility Development and shop equipment supplier expertise as required.
4. Develop/document the new staging strategy/procedure, with new roles and responsibilities.
5. Establish new/adjusted repair/rebuild time requirement targets as needed.
6. Present strategy to shop employees.
7. Install the staging processes, layout, and equipment changes into the shop.
 - a. This may occur in phases (as needed).
8. Train employees to use the new procedures/equipment. Follow-up/enforce immediately.
9. Review process/equipment performance once established.
10. Establish and install any adjustments as needed.

4.0 Benefits

- Increased capacity - These concepts reduced space requirements allowing the dealer to increase the number and size of general assembly bays.
- Reduced cost – Pans/covers/sub-assemblies are staged in protected areas instead of assembly bays. Assembly bays are rescheduled with minimal parts moves, saving set-up/tear-down time/costs.
- The dealer's technicians work more effectively with more space to organize their activity. Rebuild times and target times dropped, raising standard (flat-rate) job profits.
- Increasing quality – the rebuilt components incur less incidental damage and contamination.
- Improved Operator/Technician Safety

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5.0 Resources Required

- Investment Costs Vary
 - Well-built, protected rack for pumps cost \$1000 US while heavy-duty rollout shelves cost considerably more, depending on size, weight requirements, and supplier.
- Facility
 - Staging space (re-organized) to locate it in CRA or protected area.
- Support Equipment
 - Staging racks, carts, stands for incoming components.
 - Staging racks, carts, stands with contamination protection for out-going components.
 - Heavy duty roll-out head rack/shelves (providing crane access) can be used for larger components, or standard racks and pallet can be used with lift truck storage and retrieval
 - Standard and custom designed carts are used for shafts, pistons, rods, liners, etc. These are used for staging as well as transport.
 - Floor pallets typically consume lots of space and their use is discouraged.
- People
 - Establish current staging concerns with team.
 - Establish process, layout, and equipment improvements through team.
- Training
 - Define, establish, and enforce the new process and duties with the production team.

6.0 Supporting Attachments / References

Shown below are examples of proper parts staging.



Crankshaft rebuild area - Shafts are protected and aisle accessible for retrieval.



Fuel system rebuild area - Incoming groups on bottom shelf, outgoing groups on top shelf (protected).

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General assembly area - General parts are easily moved (wheels on rack) and located in a specific parking area.

7.0 Related Best Practices

None.

8.0 Acknowledgements

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